

Rack Inlet Temperature Prediction Based on Deep Learning

Yu-Hsuan Peng
Thermal Design Laboratory
Inventec Corporation
Taoyuan, Taiwan
peng.peteryh@inventec.com

Chien-Ming Lee
Thermal Design Laboratory
Inventec Corporation
Taoyuan, Taiwan
lee.jamescm@inventec.com

Kai-Yang Tung
Thermal Design Laboratory
Inventec Corporation
Taoyuan, Taiwan
tung.sean@inventec.com

Rongshun Chen
Department of Power
Mechanical Engineering
National Tsing Hua University
Hsinchu, Taiwan
rchen@pme.nthu.edu.tw

Abstract—Improving the cooling efficiency of a data center becomes important due to the cooling power reaches one-half of total power budget in operating the data center. The major cause of inefficient cooling is the poor airflow management resulting in cold air bypass and hot air recirculation. Basically the architecture of airflow management includes the separation of cold aisle and hot aisle and raised floor with perforated tile. The cold air is generated by the Computer Room Air Conditioning (CRAC) unit and is delivered to the server rack through the perforated floor. To avoid hot air recirculation, an active tile is used to enhance the airflow in a specific location where the flow resistance is larger than the others. The priority of the airflow control is to eliminate hot spot owing to the hot air recirculation such that the control strategy is conservative. Recently many studies pay attention to real-time temperature prediction by utilizing airflow model to improve the control performance of the airflow. However the derivation of airflow model suffers the difficulty from the airflow measurement. Therefore this research aims at the development of deep learning model for predicting the inlet temperature distributions in real time for use in control strategies in order to improve energy usage dynamically. A rack tested system was established for the validation of the temperature prediction by using deep learning. In this paper we define the workload schedule of the rack to collect the training data and the GRU recurrent neural network model is used to develop the temperature prediction model. The experimental results show that the prediction model can predict the inlet temperature at a root mean square error of 0.29°C.

Keywords—energy efficiency, hot spot, temperature prediction, deep learning, recurrent neural network

I. INTRODUCTION

A data center composes of a huge number of servers, network equipment and cooling infrastructures. It consumes significant electrical energy due to its increasing demand and power density for providing the services of social media, cloud computing and storage. Statistics indicate that over 2% of global electricity will be used to power data centers worldwide by 2030 [1]. According to the research [2], cooling accounts for 40-50% of the total energy consumption. Therefore, thermal management to improve cooling efficiency becomes important to reduce energy costs in the design of the data center.

Based on the heat exchange medium used to remove the heat from the servers, the cooling solution can be divided into three categories, air cooling, liquid cooling and immersion cooling [3]. Air cooling is the most common method used in the data center. It is circulated by the air mover and then passes through the servers to maintain their temperature under the threshold. The cold air is generated by the Computer Room Air Conditioning (CRAC) units and

further moved into the servers by the fans. Eventually, the heated hot air returns to the CRAC to complete the airflow circulation. The main issue of air cooling in the data center is the mixing of supplying cold air and returning hot air [4], resulting in inefficient air cooling and wasting electricity.

Generally, the air mixing comes from hot air recirculation and cold air bypass. The intuitive approach to reducing air mixing is to physically separate the airflow. Hence aisle containment solutions have been proposed to improve airflow efficiency [5, 6]. Moreover, literatures [7, 8] presented the raised floor with active tile architectures to improve the cooling performance. On the other hand, data center facilities operators care about the problem of downtime more than energy efficiency. The Uptime Institute reported that four in 10 outages cost between 100,000 USD and 1 million USD and about one in six costs over 1 million USD [9]. To prevent the downtime of the data center from overheating, the control of air cooling used in the data center is conservative. The control system preset higher airflow supply to the servers than it needs. As a consequence, cold air leakage occurs in the situation of the light loading of the data center.

To provide just enough cold airflow to the servers, dynamic fan control strategies were proposed for better airflow management [10], aiming at the balance between the supply and the demand of the cold airflow to the servers. To date, existing DCs cooling systems rely on the reactive control scheme. For example, in the case of the inlet temperature monitoring, the control action is reacted according to the variation of the server inlet temperature. However, the temperature may not be able to respond to the control action of the cooling fan timely. Therefore airflow model has been constructed for real-time temperature prediction [11]. Furthermore, time series analysis and machine learning-based prediction [12] is the alternative approach to avoiding sophisticated airflow modeling.

As stated above, it is believed that temperature prediction is an important key to better airflow efficiency. Therefore this paper proposes a system and method for inlet temperature prediction by using deep learning. A rack testbed was established for validation. The rest of this paper is organized as follows. In the next section, we describe the detail of the rack testbed system. The methodology of deep learning for temperature prediction is presented in Section III. Experimental evaluation and results are described in Section IV and the conclusion is given in Section V.

II. SYSTEM

In this paper, a rack testbed was first completed, and an automatic data collection system was developed based on a

42U standard rack model. The rack testbed is used to simulate the server in a real raised floor data center, in which a cold aisle is well contained. The system schematic is shown in Fig. 1. The rack testbed, which consists of dummy servers, server rack, and a cold chamber with an active tile fan. In addition, there is a movable door to enable researchers to enter the controlled environment easily. The blue area in front of the rack is the cold aisle. Except for the vents below, all possible air inlets are isolated by acrylic sheets. The rear of this structure is the hot aisle, which is the hot air removal direction.

Also, a wireless transmission and sensing system was used in this study to adjust the server statement without disturbing the flow field, and the numerous data generated from the operating server will transmit to the host PC. Users can observe and control various information and parameters through the graphical user interface (GUI). The establishment of this simulation system is conducive to the collection of subsequent training data and provides validation of our prediction models.

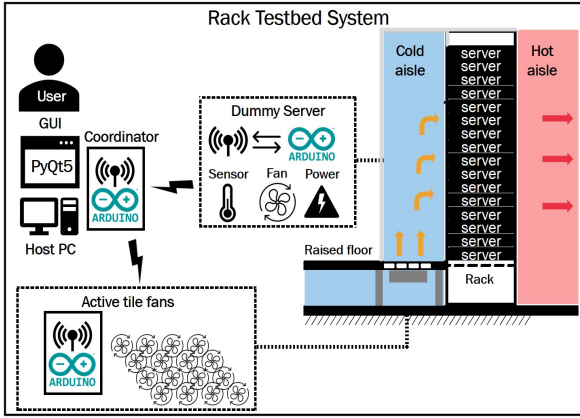


Figure 1. System Structure of rack testbed

A. Raised floor environment

The rack environment in this paper is set up of a raised floor data center computer room. The raised floor computer room is used by most existing data centers in operation. Its main advantage is to isolate the influence of hot and cold aisles. As Fig. 2 shows below, the raised floor is to set up a platform on the ground to place the server racks through the tripod and use the space between the raised floor and the ground as the transmission of cooling fluid. The aisle supplies cold air from the cooling system to the target server rack and then supplies it upward to the server through the pores of the perforated tiles.

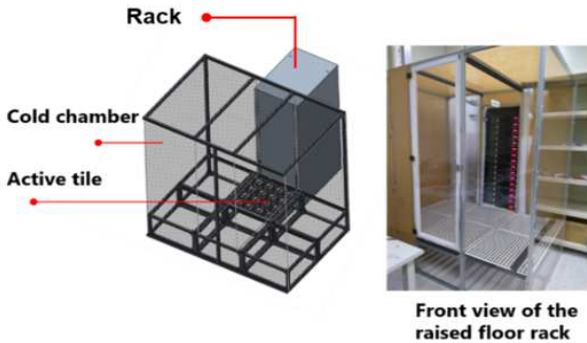


Figure 2. Raised floor room of our testbed

B. Dummy Server

In this paper, we use dummy servers instead of real servers, which can be used to simulate the physical characteristics of real servers during operation, such as temperature distribution, heat dissipation, or flow field changes. The schematic diagram of the internal structure of the dummy server is shown in Fig. 3 below. Using dummy servers can significantly reduce the complexity and cost of experiments when a large number of servers are required. The dummy server includes a chassis body, a heat generator, a wire-less sensing and control module, a fan set, and a power supply. The heat generator simulates the heating condition of the core processing chip of a real server. The fan set provides the cold air required for the cooling of the server, and the chassis is the same size as a real server that it has a similar impedance to a real server when installed inside the rack.

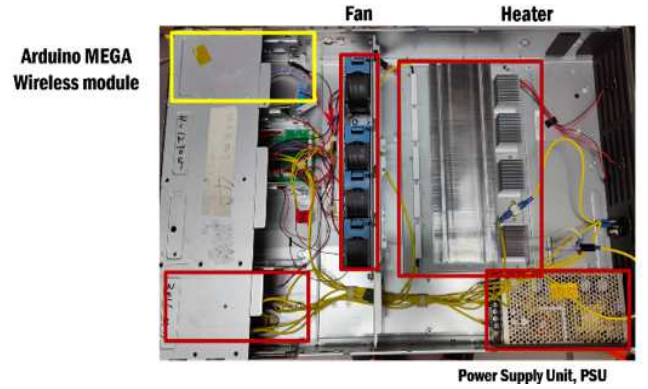


Figure 3. Top view of a dummy server

III. METHODOLOGY

The temperature prediction of the data center can be divided into conditional temperature prediction and time series temperature prediction. Among them, conditional temperature prediction considers conditions such as the state of the prediction node and the surrounding environment. In 2011, Wang [13] et al. proposed a method for predicting conditional temperature. In the time series temperature prediction, since the temperature of the server and the computing power consumption change with time, specific characteristics are extracted from the historical data of the data center, such as the current server load, temperature, and surrounding factors that may affect the temperature. For most data centers, the allocated data processing changes periodically over time, and in a closed computer room environment, the generation and dissipation of heat is a gradual change process, so this study selects time series prediction as the method of temperature prediction.

A. Time series prediction in datacenter

There are many different ways to achieve time series prediction, such as with the Autoregressive Integrated Moving Average Model or through deep learning modeling, such as using Long Short-Term Memory and The improved recurrent neural network (RNN) model has its own advantages and disadvantages, and this study uses Gated Recurrent Unit (GRU) as the development of the time series prediction model. In this study, the goal of predicting the server inlet temperature is to construct a GRU model from the single-step prediction of multivariate factors.

B. Neural network model structure

This section will introduce the neural network model used in this study, including the principles of RNN and GRU model, as well as the inlet temperature prediction model designed by GRU. GRU is a variant of RNN, it contains two control valves inside, namely the reset valve and the updated valve, which are responsible for the storage and use of memory items, so that each cycle unit can adapt to capture correlations across different time horizons. The GRU model structure is shown in Fig. 4 below, where X_t is the input feature at a specific time t , and N represents the input time series length. The input features are placed between $[0, 1]$ after pre-processing, and the dynamic changes in the time series of the inlet temperature-related feature data are processed through a layer of GRU layer, tanh function, and a layer of full connection layer (Full Connect, FC). The model's output is linearly amplified to obtain the output value Y_t .

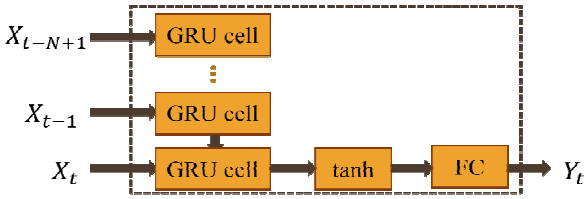


Figure 4. The structure of the GRU model in this study

In this study, Tensorflow and Keras are used to establish the model structure of the entire recurrent neural network, and the collected data is used to train the model and establish a prediction model. In order to obtain a complete inlet temperature distribution, we use 16 corresponding numbered prediction models, the inlet temperature of each server is predicted respectively, to obtain a full inlet temperature profile, and the structure diagram is shown in Fig. 5 below.

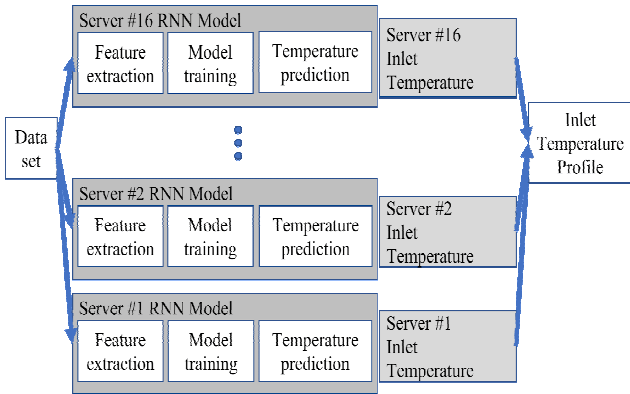


Figure 5. Rack inlet temperature model structure diagram

C. Feature Selection

The features used in this study include the server inlet temperature, outlet temperature, core sensor temperature, heating power, server fan speed, circulation fan speed, and ambient temperature, measured by LM35 and thermocouple.

The reference feature diagram is shown in Fig. 6 below. The characteristics of the target server are labeled with N , while the server information above the target is labeled with

$N+1$, and below the target is labeled with $N-1$. However, we found in subsequent experiments that adding information from nearby servers did not improve predictions. The reason may be the inlet and outlet temperature of the server itself is enough to reflect the influence of the surrounding server on it. Adding more information will make the model overfitting, and the performance of the model on the test set will decrease.

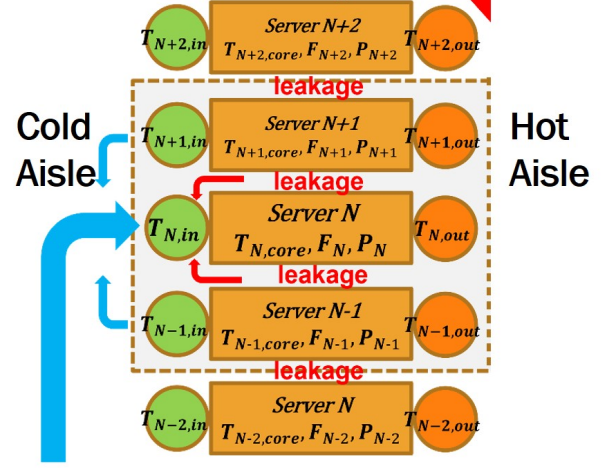


Figure 6. Schematic diagram of thermal-related feature

D. Training data collection

When there are a large number of servers, there are many permutations and combinations of their operation modes. In this study, in order to obtain the feasibility of the experiment and simplify the complexity of the characteristic data, 16 servers are operated in groups [14], the servers are divided into 4 groups, and 16 load patterns are used to measure and analyze the inlet temperature distribution.

As Fig. 7 shows below, the load in this study are divided into 16 different patterns, Case A, Case B, Case C, Case D, and Case E. There are different distributions in each pattern, the load is divided into the high load (about 600 watts), marked in red and the low load (about 100 watts), marked in green. The rack is fully loaded at 9,600 watts in total. The complete data collected according to the patterns in Fig. 7 has a total of about 80,000 seconds of training data.

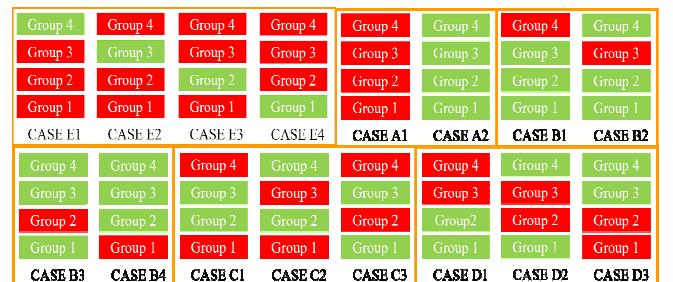


Figure 7. Schematic diagram of rack load patterns

IV. RESULT

This chapter shows the result of our rack testbed system and prediction model. The previously collected data are divided into training sets and validation sets, and a test set is a random combination of server load patterns.

A. Rack Testbed System

In this experiment, 16 dummy servers are installed in the server rack, and they are operated in the rack testbed to observe their cooling performance. The core temperature of each server is set at 50°C, the result is shown in Fig. 8 below. We can observe that except for the servers numbered 1 and 9, the core temperature of the other servers has an error of about 0.5°C from the set point. In addition, there is no obvious hot spot in the inlet temperature.

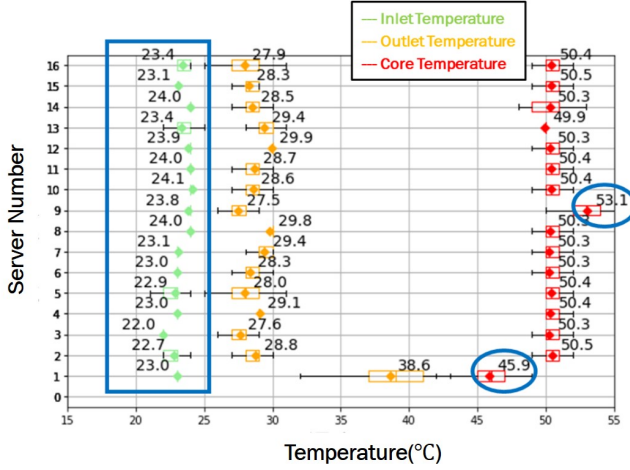


Figure 8. Temperature measurement in rack testbed system

B. Training, validation and testing

The data we collected earlier were divided into training and validation sets in a ratio of 8:2, and a new data set was created by randomly switching load patterns for testing. The error results of each server number are repeatedly tested five times to obtain the average value. Fig. 9 shows 4 examples of prediction results and the ground truth temperature measurement. The prediction horizon is set to be 10 seconds, the green line is the prediction result, and the blue line is the real data. It can observe that our prediction results well match the ground truth measurements of server inlet. The prediction results of test data set show that the average root mean square error of 16 servers is 0.29 °C, and the average maximum error is 3.9 °C.

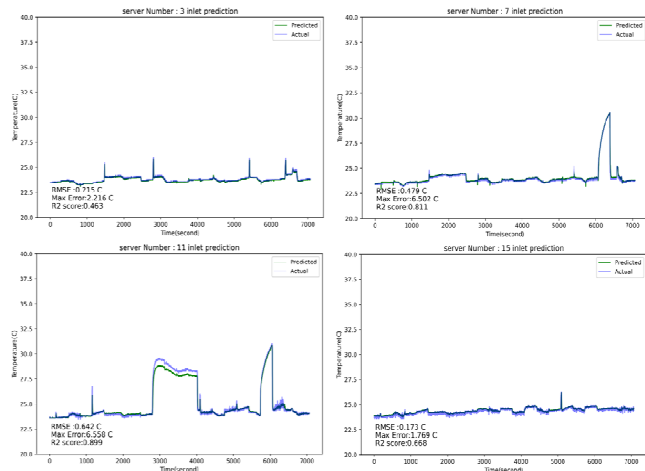


Figure 9. Server inlet temperature prediction result

V. CONCLUSION

In order to improve the efficiency of airflow control, this paper proposes the inlet temperature prediction of the rack based on the deep learning model. A rack testbed system is

established for the validation of the inlet temperature prediction. It includes a rack equipped with 16 dummy servers, a cold aisle containment for the separation of cold air and hot air, and the raised active floor tile for enhancing the airflow. The deep learning model is developed by the GRU recurrent neural network. To train the deep learning model, the workload of the rack is scheduled and the relative training data is obtained through the wireless sensor network. The experiments show that the GRU model can predict the future inlet temperature of each server according to the historical information. Using the random arrangement workload patterns as the testing data set, the average root mean square error and the average maximum error of the inlet temperature of each server are 0.29°C and 3.9°C respectively.

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